SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

---

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

---

QUESTIONS: 05 **Total Marks:50**

Annexure A

Constraint-Based Problem Solving

Criteria	Problem Understanding *(2.5)*	Constraint Analysis *(2.5)*	Solution Development *(2.5)*	Application of Concepts *(1.5)*	Justification *(1)*	***Total(10)***
Weightage	*25%*	25%	25%	15%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I M MEGHANADH(192325026) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
MEGHANADH

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Q1. Dataset: Hospital Patient Records

PID	Symptoms Observed
P1	Fever, Cough
P2	Fever, Headache
P3	Cough, Headache
P4	Fever, Cough, Headache
P5	Fever, Fatigue

Question:
Apply association rule mining with minimum support = 50% and confidence = 70%, under the constraint that all generated rules must include **"Fever"**. Identify the valid association rules and justify why they satisfy the given constraints.

Q2. Dataset: Online Learning Platform

Student ID	Courses Enrolled
S1	Python, SQL
S2	Python, Data Science
S3	SQL, Machine Learning
S4	Python, SQL, Data Science
S5	Python, Machine Learning

Question:
Using the dataset, generate frequent itemsets with minimum support = 40%, applying the constraint that each itemset must contain **at least two courses**. Explain how the constraint helps in pruning candidate itemsets during mining.

Q3. Dataset: Library Borrowing Records

Transaction ID	Books Borrowed

T1	Data Mining, Python Programming
T2	Data Mining, Database Systems
T3	Python Programming, Database Systems
T4	Data Mining, Python Programming, Database Systems
T5	Data Mining, Artificial Intelligence

Question:
Apply the Apriori algorithm with the constraint that the **maximum itemset size is 3**. Identify the frequent itemsets and explain how the constraint reduces computational complexity during candidate generation.

Q4. Dataset: University Course Hierarchy

Student ID	Courses Taken
S1	Engineering, Computer Science, Machine Learning
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

Question:
Apply multilevel association rule mining with the constraint that rules should be generated only at the **"Computer Science"** level. Explain how the course hierarchy affects the discovery of association rules.

Q5. Dataset: Insurance Customer Records

CID	Age Group	Income Level	Policy Purchased
C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Question:
Apply multidimensional association rule mining with the constraint **Age Group = "Young"**. Identify meaningful patterns involving income level and policy type, and explain how these patterns can support targeted marketing strategies

Q1. Hospital Patient Records

Given Dataset

Patient	**Symptoms**
P1	Fever, Cough
P2	Fever, Headache
P3	Cough, Headache
P4	Fever, Cough, Headache
P5	Fever, Fatigue

Minimum Support = **50%**

Minimum Confidence = **70%**

Constraint: Every generated rule must include **Fever**.

Step 1: Calculate Support

There are 5 transactions.

Itemset	**Count**	**Support**
Fever	4	4/5 = **80%**
Cough	3	3/5 = **60%**
Headache	3	3/5 = **60%**
Fatigue	1	1/5 = **20%**
Fever, Cough	2	2/5 = **40%**
Fever, Headache	2	2/5 = **40%**
Fever, Fatigue	1	1/5 = **20%**

Step 2: Apply Minimum Support

Minimum support is 50%.

Therefore:

- **Fever → 80%**, frequent
- **Cough → 60%**, frequent
- **Headache → 60%**, frequent
- Fever + Cough → 40%, eliminated
- Fever + Headache → 40%, eliminated
- Fever + Fatigue → 20%, eliminated

Step 3: Generate Rules

Since every rule must contain **Fever**, possible rules include:

- Fever → Cough
- Cough → Fever

- Fever → Headache
- Headache → Fever

However, all these rules have support below 50% because their combined itemsets occur only twice.

For example:

Fever → Cough

$$Support(F, everCough) = \frac{2}{5} = 40\%$$

$$Confidence(Fever \rightarrow Cough) = \frac{Support(F, everCough)}{Support(Fever)} = \frac{40}{80} = 50\%$$

Confidence is also below 70%.

Final Answer

There are no valid non-trivial association rules satisfying both minimum support = 50% and minimum confidence = 70% while including Fever.

The constraint **Fever** reduces the search space because only itemsets containing Fever need to be considered. The minimum-support constraint eliminates Fever-Cough, Fever-Headache and Fever-Fatigue combinations.

Q2. Online Learning Platform

The dataset contains five students. The minimum support is **40%**, and every itemset must contain **at least two courses**.

Dataset

Student	Courses
S1	Python, SQL
S2	Python, Data Science
S3	SQL, Machine Learning
S4	Python, SQL, Data Science
S5	Python, Machine Learning

Minimum support:

$$40\% \times 5 = 2$$

Therefore, an itemset must occur in **at least 2 students**.

Step 1: 2-Course Itemsets

Itemset	Students	Support
Python, SQL	S1, S4	2/5 = **40%**
Python, Data Science	S2, S4	2/5 = **40%**

Itemset	Students	Support
Python, Machine Learning	S5	1/5 = 20%
SQL, Data Science	S4	1/5 = 20%
SQL, Machine Learning	S3	1/5 = 20%
Data Science, Machine Learning	None	0%

Frequent Itemsets

Therefore:

1. **{Python, SQL} → 40%**
2. **{Python, Data Science} → 40%**

Step 2: 3-Course Itemsets

Possible:

{Python, SQL, Data Science}

Only S4 has all three.

$$Support = \frac{1}{5} = 20\%$$

Therefore, it is eliminated.

Other three-course combinations also have support below 40%.

Final Answer

The frequent itemsets satisfying both constraints are:

Frequent Itemset	Support
{Python, SQL}	**40%**
{Python, Data Science}	**40%**

How the Constraint Helps Pruning

The **minimum two-course constraint** immediately removes all single-course itemsets from consideration. The minimum support constraint then removes combinations appearing in fewer than two students.

Thus, candidate generation becomes smaller and the number of support calculations is reduced, improving mining efficiency.

Q3. Library Borrowing Records

The question asks to apply **Apriori** with a maximum itemset size of **3**. The source does **not specify a minimum-support threshold**, so a unique set of "frequent" itemsets cannot technically be determined.

Therefore, I will show the itemsets and their support so you can apply whichever minimum-support threshold your faculty expects.

Dataset

Transaction	Books
T1	Data Mining, Python Programming
T2	Data Mining, Database Systems
T3	Python Programming, Database Systems
T4	Data Mining, Python Programming, Database Systems
T5	Data Mining, Artificial Intelligence

Step 1: 1-Itemsets

Item	Count	Support
Data Mining	4	**80%**
Python Programming	2	**40%**
Database Systems	3	**60%**
Artificial Intelligence	1	**20%**

Step 2: 2-Itemsets

Itemset	Count	Support
Data Mining, Python Programming	2	**40%**
Data Mining, Database Systems	2	**40%**
Data Mining, Artificial Intelligence	1	**20%**
Python Programming, Database Systems	2	**40%**
Python Programming, Artificial Intelligence	0	**0%**
Database Systems, Artificial Intelligence	0	**0%**

Step 3: 3-Itemsets

The possible meaningful 3-itemset is:

{Data Mining, Python Programming, Database Systems}

It occurs in T4 only.

$$Support = \frac{1}{5} = 20\%$$

Other 3-itemsets have 0% support.

If Minimum Support = 40%

The frequent itemsets would be:

1-itemsets

- {Data Mining} → 80%
- {Python Programming} → 40%
- {Database Systems} → 60%

2-itemsets

- {Data Mining, Python Programming} → 40%
- {Data Mining, Database Systems} → 40%
- {Python Programming, Database Systems} → 40%

3-itemsets

- None, because the only 3-itemset has 20%.

Effect of Maximum Size = 3

The constraint prevents Apriori from generating itemsets containing **4 or more books**.

Therefore, the algorithm only needs to generate:

$$L_1 \rightarrow L_2 \rightarrow L_3$$

and stops after size 3.

This reduces:

- Number of candidates
- Support calculations
- Memory requirements
- Computational time

Final Answer

The maximum-itemset-size constraint significantly reduces computational complexity because candidate generation stops at three-item combinations instead of exploring larger combinations.

Q4. University Course Hierarchy

The dataset is:

Student	**Courses**
S1	Engineering, Computer Science, Machine Learning
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

The constraint is that rules should be generated **only at the Computer Science level**.

Step 1: Frequency of Computer Science

Computer Science occurs in:

- S1
- S3
- S4

Therefore:

$$Support(CS) = \frac{3}{5} = 60\%$$

Step 2: Relationship with Engineering

Engineering occurs in all five students.

Computer Science + Engineering occurs in:

- S1
- S3
- S4

Therefore:

$$Support(C, SEngineering) = \frac{3}{5} = 60\%$$

Rule 1: Computer Science → Engineering

$$Confidence = \frac{Support(C, SEngineering)}{Support(CS)}$$

$$= \frac{60\%}{60\%} = 100\%$$

So:

Computer Science → Engineering

has:

- Support = **60%**
- Confidence = **100%**

Rule 2: Engineering → Computer Science

$$Confidence = \frac{60\%}{100\%} = 60\%$$

So:

Engineering → Computer Science

has:

- Support = **60%**
- Confidence = **60%**

Step 3: Relationship with Machine Learning

Computer Science + Machine Learning occurs in:

- S1
- S4

Therefore:

$$Support(C, SML) = \frac{2}{5} = 40\%$$

Computer Science → Machine Learning

$$Confidence = \frac{40\%}{60\%} = 66.67\%$$

Machine Learning → Computer Science

Machine Learning occurs in S1 and S4:

$$Support(ML) = \frac{2}{5} = 40\%$$

Therefore:

$$Confidence = \frac{40\%}{40\%} = 100\%$$

Important Rules at Computer Science Level

Rule	Support	Confidence
Computer Science → Engineering	**60%**	**100%**
Engineering → Computer Science	**60%**	**60%**
Computer Science → Machine Learning	**40%**	**66.67%**
Machine Learning → Computer Science	**40%**	**100%**

Effect of Course Hierarchy

Multilevel association mining allows relationships to be discovered at a particular level of a hierarchy.

Here, restricting mining to the **Computer Science level** avoids generating unrelated rules involving Information Technology.

The results show that students taking Computer Science also take Engineering, while Machine Learning students in this dataset always take Computer Science.

Q5. Insurance Customer Records

The question requires multidimensional association-rule mining with the constraint:

Age Group = Young.

Step 1: Select Young Customers

Original data:

Customer	Age	Income	Policy
C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

After applying the constraint **Age Group = Young**:

Customer	Income	Policy
C1	High	Health Insurance

Customer	Income	Policy
C3	Low	Travel Insurance
C5	Medium	Vehicle Insurance

There are **3 Young customers**.

Step 2: Identify Patterns

High Income → Health Insurance

High-income young customer:

- C1

$$Support = \frac{1}{3} = 33.33\%$$

Low Income → Travel Insurance

Low-income young customer:

- C3

$$Support = \frac{1}{3} = 33.33\%$$

Medium Income → Vehicle Insurance

Medium-income young customer:

- C5

$$Support = \frac{1}{3} = 33.33\%$$

Pattern Table

Income Level	Policy	Count	Support
High	Health Insurance	1	33.33%
Low	Travel Insurance	1	33.33%
Medium	Vehicle Insurance	1	33.33%

Interpretation

The constrained dataset suggests:

- **Young + High Income → Health Insurance**
- **Young + Medium Income → Vehicle Insurance**
- **Young + Low Income → Travel Insurance**

However, each combination appears only once. Therefore, these should be treated as **observed patterns rather than strong association rules**.

Marketing Application

The insurance company could use these observations for targeted marketing:

- Young customers with **higher income** could be shown health-insurance campaigns.
- Young customers with **medium income** could receive vehicle-insurance offers.

- Young customers with **lower income** could be presented with affordable travel-insurance options.

The **Age = Young constraint** reduces the dataset before mining, making the analysis more focused on the desired customer segment.